Quasi-amorphous crystals: Supplemental Material

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a. Fundamental and elastic domains. Consider the energy density function $\varphi(\mathbf{C})$ with tensorial symmetry $GL(2,\mathbb{Z}) = \{\mathbf{m},\, m_{IJ} \in \mathbb{Z},\, \det(\mathbf{m}) = \pm 1\}$. We refer to the restriction of such periodic energy density to the minimal periodicity (fundamental) domain as $\varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$. Here $\tilde{\mathbf{C}} = \mathbf{m}^T\mathbf{C}\mathbf{m}$ is the projection of a general metric tensor $\mathbf{C}$ into the domain $\mathscr{D}$ and $\mathbf{m}$ is the corresponding unimodular integer valued matrix that performs the projection while ensuring that $\varphi(\mathbf{C}) = \varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$. The actual configuration of the fundamental domain $\mathscr{D}$ is well known [1–4]

$$\mathscr{D} = \left\{ 0 < C_{11} \leq C_{22}, \quad 0 \leq C_{12} \leq \frac{C_{11}}{2} \right\}, \tag{1}$$

where $\det \mathbf{C} = 1$. Given a generic metric $\mathbf{C}$, the task of finding a unimodular matrix $\mathbf{m}$ ensuring that $\tilde{\mathbf{C}} \in \mathscr{D}$ (and therefore $\varphi(\mathbf{C}) = \varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$) can be formulated as a recursive algorithm which is also well known (Lagrange reduction) [2, 3]. Specifically, if we define the matrices

$$\mathbf{m}_1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},\ \ \mathbf{m}_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},\ \ \mathbf{m}_3 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \tag{2}$$

and start with an assumption that $\mathbf{m} = \mathbb{I}$, we can proceed through the following steps: (i) if $C_{12} < 0$, change sign of C_{12} using the mapping $\mathbf{m} \to \mathbf{m}\mathbf{m}_1$; (ii) if $C_{22} < C_{11}$, swap these two components using the mapping $\mathbf{m} \to \mathbf{m}\mathbf{m}_2$; (iii) if $2C_{12} > C_{11}$, set $C_{12} = C_{12} - C_{11}$ using the mapping $\mathbf{m} \to \mathbf{m}\mathbf{m}_3$. Note that the action of $\mathbf{m}_1$ is a reflection that ensures an acute angle between two lattice vectors of the square lattice $\mathbf{e}_i$ where $i = 1, 2$; the action of the matrix $\mathbf{m}_2$ is also a reflection as it swaps two lattice vectors $\mathbf{e}_i$. Both of these operations belong to the point group and propagate the metric only inside the corresponding 'elastic domain' (Ericksen-Pitteri neighborhood) composed of the four copies of the minimal domain $\mathscr{D}$ [4, 5]. Instead, the mapping defined by matrix $\mathbf{m}_3$ brings the metric outside the 'elastic domain' and therefore represents a quantized analog of the macroscopic plastic strain.

b. Elastic energy density. While the single-period Landau potential $\varphi_{\mathscr{D}}(\tilde{\mathbf{C}})$ can be constructed using the classical Cauchy-Born approach, see for instance [5, 6], in this paper we used, for simplicity, the phenomenological expression proposed in [2]. Specifically, the elastic energy density is represented as a sum of two terms:

$$\varphi(\tilde{\mathbf{C}}) = \varphi_0\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right) + \varphi_v(\det \tilde{\mathbf{C}}), \tag{3}$$

where φ_0 accounts for contributions due to shear while φ_v penalizes volumetric deformations. The $GL(2,\mathbb{Z})$ invariant shear term is chosen in the form a sixth-order polynomial which is the minimal requirement ensuring stress continuity over the whole configurational space [1, 2]. It depends on a single parameter β allowing one to associate ground state either with square or triangular (hexagonal) lattice

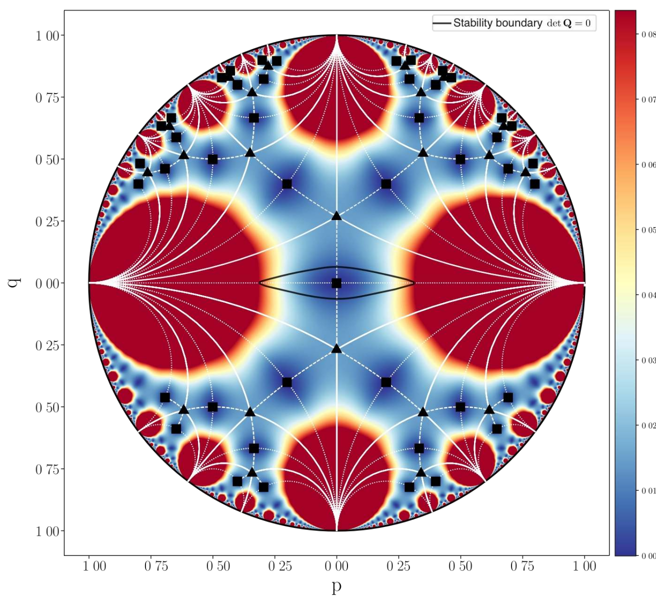

FIG. 1. Strain-energy density in the configurational space C_{11}, C_{22}, C_{12} with $\det \mathbf{C} = 1$ stereographically projected on the Poincaré disk. Thin white lines represent the $GL(2,\mathbb{Z})$ tessellation of the a Poincaré disk into equivalent periodicity domains. Colors represent energy levels. Elastic stability boundary given by (10) is shown by the thick black contour surrounding the reference state. Small black squares and triangles correspond to equivalent square and triangular lattices, respectively.

$$\varphi_0\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right) = \beta\psi_1\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right) + \psi_2\left(\frac{\tilde{\mathbf{C}}}{(\det \tilde{\mathbf{C}})^{1/2}}\right). \tag{4}$$

The requirement of stress continuity across the whole periodic energy landscape specifies the functions ψ_1 and ψ_2

$$\psi_1 = I_1^4 I_2 - \frac{41}{99} I_2^3 + \frac{7}{66} I_1 I_2 I_3 + \frac{1}{1056} I_3^2, \tag{5}$$

$$\psi_2 = \frac{4}{11} I_2^3 + I_1^3 I_3 - \frac{8}{11} I_1 I_2 I_3 + \frac{17}{528} I_3^2. \tag{6}$$

where we used the known hexagonal invariants of the metric tensor

$$I_1 = \frac{1}{3}(\tilde{C}_{11} + \tilde{C}_{22} - \tilde{C}_{12}), \tag{7}$$

$$I_2 = \frac{1}{4}(\tilde{C}_{11} - \tilde{C}_{22})^2 + \frac{1}{12}(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^2, \tag{8}$$

$$I_3 = (\tilde{C}_{11} - \tilde{C}_{22})^2(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12}) - \frac{1}{9}(\tilde{C}_{11} + \tilde{C}_{22} - 4\tilde{C}_{12})^3. \tag{9}$$

We select $\beta = -1/4$ to ensure that global energy minimizers correspond to square lattices. The volumetric part of the energy density, which primarily influences the fine structure of the dislocation cores and also controls the formation of voids, is assumed to be of a generic form preventing infinite compression $\varphi_v(s) = \mu(s - \log(s))$. To keep in our numerical experiments the strain field close to the surface $\det \mathbf{C} = 1$, we adopted a sufficiently high value of the bulk modulus, $\mu = 5$.

The isochoric part of the energy landscape is illustrated in Fig. 2 of the main paper. Here we also show the same Landau-type potential in Fig. 1, but now in the level set representation. Dark blue regions correspond to the equivalent square energy wells. Red domains show effectively inaccessible regions where the energy is very high (truncated). Relatively low energy valleys colored in yellow correspond roughly to two simple shear plastic 'mechanisms' available in square lattices.

c. Internal length scale. The lack of convexity of the potential is a property which the MTM shares with other similar Landau-type continuum theories. In the MTM approach the necessary regularization is introduced by bringing into the model an internal scale h through explicit spatial discretization. If L is the linear size of the macroscopic domain and N^2 is the number of nodes in the mesoscopic grid, then the parameter $h = L/N$ introduces elastic finite elements imitating mesoscopic aggregates of atomic particles and defines in this way the cutoff beyond which the deformation is considered homogeneous (affine). The corresponding small parameter is $\delta = h/L = 1/N$. Another small parameter in the problem is $\epsilon = a/L$ where a is the interatomic distance and we are interested in the double limit: $\epsilon \to 0$ and $\delta \to 0$ while $\delta \gg \epsilon$. In the MTM approach we first implicitly perform the limit $\epsilon \to 0$ and recover in this way the continuum constitutive response using the Cauchy-Born rule. Then instead of performing the classical $\delta \to 0$ and obtaining the conventional scale-free continuum theory, we preserve in the theory a small but finite value of the parameter δ. In this way we effectively account for finite size effects. In particular, the size of the dislocation cores is overestimated while the number of dislocations is underestimated. However, the resulting approach preserves the basic nature of both long-range and short-range interactions of dislocations at least when they are away from the boundaries. For instance, the blown up dislocations will still nucleate and self-lock adequately.

In our numerical code we used dimensionless parameters $\tilde{h} = h/b = 1$ and $\tilde{L} = L/b = 1/N$, where the b is the size of the mesoscopic particle. For the choice of the scale b it is natural to require that the Cauchy-Born energy density computed for a lattice fragment with scale b exhibits a high level of periodicity within the range of strains reached in a particular numerical experiment. We have checked that in our case the assumption that $b \sim 10a$ is sufficient.

d. Numerical method. Numerical implementation of the MTM approach reduces to solving an elastic finite element problem. The goal is to follow the displacements of the network of discrete nodes labeled by integer-valued coordinates $I = 1, ..., N^2$.

Specifically, we assume that each of the elements is a deformable triangle and employ the standard linear 3-node elements [7, 8]. The displacement field inside each of the elements, represented in dimensionless form, can be written in the form $\mathbf{u}(\mathbf{z}) = \sum_a \mathcal{N}^a(\mathbf{z}; h)\mathbf{u}^a$, where $\mathcal{N}^a(\mathbf{z}; \mathbf{h})$ are compactly supported linear shape functions, $\mathbf{u}^a$ are the are the nodal displacements and summation is assumed over repeated indexes; the interpolation functions for each element are defined in terms of a local dimensionless coordinate system. The mesoscopic deformation gradient is then $\mathbf{F}(\mathbf{z}; h) = \mathbb{I} + \mathbf{u}^a \otimes \nabla\mathcal{N}^a(\mathbf{z}; h)$.

In terms of macroscopic reference coordinates, the elastic energy inside each element can be computed using the simplest quadrature scheme $w(\mathbf{x}; h) = \frac{1}{2}\varphi(\mathbf{F}(\mathbf{x}; h))J(\mathbf{x}, h)$, where J is the Jacobian of the transformation from local $(\mathbf{z})$ to global $(\mathbf{x})$ coordinate system. Note that in view of our assumptions the energy of a deformed triangular element depends on three parameters: the deformed lengths of the bonds and the deformed value of the angle between them.

Finding a solution of an elastic problem implies local minimization of the energy $W = \int_\Omega w(\mathbf{x}; h)d\mathbf{x}$, which is prescribed on a triangulated domain Ω. The conditions of mechanical equilibrium take the form $\nabla \cdot \mathbf{P} = 0$ where we introduced the first Piola-Kirchhoff stress tensor $\mathbf{P} = 2\mathbf{F}\mathbf{m}\mathbf{\Sigma}\mathbf{m}^T$, with $\mathbf{\Sigma} = \begin{bmatrix} \frac{\partial w}{\partial \tilde{C}11} & \frac{1}{2}\frac{\partial w}{\partial \tilde{C}12} \\ \frac{1}{2}\frac{\partial w}{\partial \tilde{C}12} & \frac{\partial w}{\partial \tilde{C}22} \end{bmatrix}$. In the finite element representation these equations reduce to

$$\frac{\partial W}{\partial \mathbf{u}^a} = \int_\Omega \mathbf{P}(\mathbf{x}; h)\nabla\mathcal{N}^a(\mathbf{x}; h)d\mathbf{x} = 0.$$

The ensuing nonlinear equilibrium problem is solved numerically using the L-BFGS algorithm [9], which constructs a positive definite approximation to the Hessian, enabling quasi-Newton steps that progressively reduce the total energy. Iterations continue until the energy change per iteration falls below a prescribed tolerance. At each iteration, starting from an approximate solution $\mathbf{w}^a$, we compute a displacement correction $\mathrm{d}\mathbf{w}^a = \mathbf{u}^a - \mathbf{w}^a$ by solving the linearized equilibrium equations via LU factorization [10, 11]:

$$K_{ij}^{ab} dw_j^b + R_i^a = 0,$$

where

$$K_{ij}^{ab} = A_{ipjq}(\mathbf{F})\frac{\partial \mathcal{N}^a}{\partial x_p}\frac{\partial \mathcal{N}^b}{\partial x_q}$$

is the global stiffness matrix and

$$R_i^a = P_{ip}(\mathbf{F})\frac{\partial \mathcal{N}^a}{\partial x_p},$$

is the residual force vector with summation implied over repeated indices. Note that we introduced the tensor of tangential elastic moduli

$$A_{iajb} = \frac{\partial^2 \varphi_{\mathscr{D}}(\tilde{\mathbf{C}})}{\partial F_{ia}\partial F_{jb}}.$$

which can be used construct the Eulerian acoustic tensor

$$Q_{ij} = F_{la}F_{mb}A_{iajb}n_l n_m,$$

where $\mathbf{n}$ is an arbitrary unit vector. The Legendre-Hadamard condition which marks the threshold of elastic instability in continuum theory can be then written in the form

$$\det(\boldsymbol{Q}) = 0. \tag{10}$$

The corresponding stability boundary is referred to in the main paper and it is illustrated for the chosen elastic energy landscape in Fig. 1.

e. Implementation of the loading conditions. To simulate an effectively infinite crystal under controlled macroscopic affine deformation, we use the method of domain replication. Starting from the computational domain Ω_0 with dimensions $L_x \times L_y$, we generate a periodic lattice of domains using translation vectors $\mathbf{t}_n = \begin{pmatrix} n_x L_x \\ n_y L_y \end{pmatrix}$, where $\mathbf{n} = (n_x, n_y) \in \{-1, 0, 1\}^2$. Let $\bar{\mathbf{F}}(\alpha)$ denote the macroscopic deformation gradient tensor at the parameter value α. Under such affine deformation, the position of replicated domains relative to the central domain Ω_0 is given by: $\mathbf{R}_n = \bar{\mathbf{F}}(\alpha) \cdot \mathbf{t}_n$, where $\mathbf{R}_n$ represents the displacement vector of the origin of the domain associated with $\mathbf{t}_n$. Then, for a node located at position $\mathbf{x}_i$ in the central domain Ω_0, its periodic image, associated with the translation vector $\mathbf{t}_n$, is at: $\mathbf{x}_i^{(n)} = \mathbf{x}_i + \mathbf{R}_n = \mathbf{x}_i + \bar{\mathbf{F}}(\alpha) \cdot \mathbf{t}_n$. Such correspondence ensures that the macroscopic deformation gradient $\bar{\mathbf{F}}(\alpha)$ is consistently applied across all periodic images of Ω_0. After establishing the positions of all nodes across the nine domains (original domain Ω_0 and its eight periodic images), we perform a Delaunay triangulation on the complete set of particles: $\mathcal{P} = \{\mathbf{x}_i : i \in \mathcal{I}_0\} \cup \{\mathbf{x}_i^{(n)} : i \in \mathcal{I}_0, n \neq \mathbf{0}\}$, where $\mathcal{I}_0$ denotes the set of particle indices in the original domain and $\mathbf{0} = (0, 0)$ corresponds to the null translation vector. Using the obtained global triangulation $\mathcal{T}$, we select the triangles that have at least one vertex belonging to the original domain Ω_0.

Note that since triangles near periodic boundaries may be counted multiple times through different domain images, we need to ensure uniqueness by mapping all vertex indices to their representatives in Ω_0 and using the sorted tuple of these representatives as a canonical identifier. Specifically, a triangle is included in $\mathcal{T}_{\text{unique}}$ only upon its first encounter, ensuring each physical triangle is counted exactly once. The resulting computational mesh $\mathcal{T}_{\text{unique}}$ is illustrated in Fig. 2, where the central domain Ω_0 (highlighted by the dark box) and the selected triangles connecting to its particles are shown. Such selection procedure ensures that: (i) Each triangle in the computational mesh is counted exactly once, eliminating redundancy from the periodic extension; (ii) All triangles connected to the original domain are included, maintaining proper connectivity across periodic boundaries; (iii) The triangulation naturally incorporates the macroscopic deformation through the deformed positions of the periodic images.

The selected set of triangles $\mathcal{T}_{\text{unique}}$ was used as the computational mesh in the MTM based numerical experiments ensuring both, compatibility with the periodic boundary conditions and consistency with the applied macroscopic deformation gradient $\bar{\mathbf{F}}(\alpha)$.

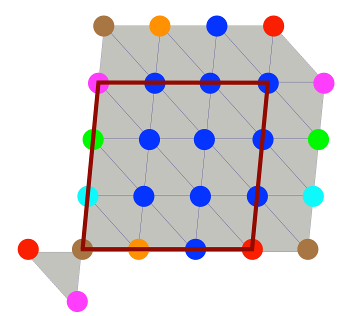

FIG. 2. Illustration of the computational mesh $\mathcal{T}_{\text{unique}}$ obtained from the periodic Delaunay triangulation. The central computational domain Ω_0 is highlighted by the dark red box. Only triangles with at least one vertex in Ω_0 are selected for the mesh (shown by the thin lines), ensuring proper connectivity across periodic boundaries while avoiding redundant counting of triangles.

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